

## MANHATTAN 6 SSW, KS, USA

WMO: 724540

Lat: 39.1027N

Lon: 96.6098W

Elev: 347

StdP: 97.23

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53974

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.1      | -13.5 | -20.9                         | 0.6 | -14.5 | -18.8 | 0.7 | -12.8 | 7.9                   | 6.1  | 7.3 | 3.7  | 1.9                   | N/A  | 0.453 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.6                   | 36.8            | 23.2 | 34.9 | 23.4 | 33.1 | 23.1 | 25.5                | 32.4 | 24.8 | 31.9 | 24.1 | 31.1 | 3.6                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.7                            | 19.3 | 29.5 | 22.8 | 18.2 | 28.8 | 21.9 | 17.3 | 28.1 | 80.4          | 32.7 | 77.2 | 31.9 | 74.2 | 31.0 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.5               | 6.6  | 5.8 | -21.0                      | 39.1 | 2.9                | 1.9 | -23.1                                              | 40.4 | -24.8      | 41.6 | -26.4      | 42.6 | -28.4      | 44.0 |    |    |
|                   |      |     | -21.7                      | 26.9 | 3.2                | 0.9 | -24.0                                              | 27.6 | -25.8      | 28.1 | -27.6      | 28.6 | -29.9      | 29.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.1   | -1.0  | 0.5   | 7.9   | 13.1  | 18.4  | 23.9  | 26.1  | 24.9  | 21.0  | 13.7  | 6.9   | 0.6  | (6)  |
| (7)  |                                                                                  | DBStd       | 10.95  | 6.61  | 6.87  | 6.49  | 5.62  | 4.88  | 3.59  | 3.66  | 3.62  | 4.68  | 5.50  | 5.97  | 6.01 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1213   | 342   | 273   | 118   | 32    | 3     | 0     | 0     | 0     | 0     | 26    | 127   | 293  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2748   | 599   | 499   | 328   | 173   | 63    | 3     | 1     | 1     | 26    | 163   | 343   | 549  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2329   | 1     | 7     | 52    | 125   | 263   | 416   | 498   | 461   | 329   | 140   | 34    | 3    | (10) |
| (11) |                                                                                  | CDD18.3     | 823    | 0     | 0     | 4     | 16    | 65    | 169   | 241   | 203   | 105   | 19    | 1     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 8817   | 0     | 2     | 44    | 172   | 599   | 1812  | 2781  | 2169  | 1030  | 198   | 12    | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3869   | 0     | 0     | 7     | 42    | 186   | 764   | 1401  | 1017  | 399   | 52    | 1     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.7    | 2.6   | 2.8   | 3.3   | 3.5   | 2.8   | 2.7   | 2.2   | 2.1   | 2.3   | 2.5   | 2.7   | 2.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 849    | 18    | 27    | 51    | 80    | 120   | 128   | 117   | 77    | 60    | 40    | 30    | (15) |      |
| (16) |                                                                                  | PrecMax     | 1319   | 40    | 83    | 119   | 235   | 334   | 259   | 393   | 216   | 135   | 143   | 126   | 93   | (16) |
| (17) |                                                                                  | PrecMin     | 619    | 2     | 2     | 0     | 24    | 28    | 9     | 47    | 19    | 11    | 5     | 1     | 1    | (17) |
| (18) |                                                                                  | PrecStd     | 161    | 10    | 20    | 31    | 43    | 80    | 58    | 77    | 42    | 33    | 33    | 35    | 26   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 18.4  | 21.8  | 27.3  | 30.7  | 32.9  | 36.9  | 39.6  | 38.9  | 35.2  | 31.2  | 25.2  | 18.1 | (19) |
| (20) |                                                                                  |             | MCWB   | 11.0  | 11.2  | 16.1  | 18.4  | 20.7  | 23.3  | 22.9  | 23.7  | 22.2  | 20.1  | 15.2  | 11.6 | (20) |
| (21) |                                                                                  | 2%          | DB     | 14.6  | 17.9  | 24.1  | 27.6  | 30.6  | 34.5  | 37.5  | 36.0  | 32.8  | 27.9  | 22.0  | 14.7 | (21) |
| (22) |                                                                                  |             | MCWB   | 7.9   | 10.2  | 14.2  | 16.6  | 20.2  | 23.8  | 23.4  | 22.9  | 22.4  | 18.1  | 14.3  | 8.9  | (22) |
| (23) |                                                                                  | 5%          | DB     | 11.6  | 14.5  | 21.4  | 25.2  | 28.8  | 32.8  | 35.2  | 34.0  | 31.0  | 25.4  | 19.2  | 12.2 | (23) |
| (24) |                                                                                  |             | MCWB   | 6.1   | 9.0   | 12.7  | 15.6  | 19.6  | 23.5  | 23.7  | 22.9  | 21.8  | 16.6  | 11.9  | 7.2  | (24) |
| (25) |                                                                                  | 10%         | DB     | 8.8   | 11.4  | 18.6  | 22.8  | 26.9  | 30.9  | 33.2  | 31.9  | 29.0  | 23.1  | 16.5  | 9.8  | (25) |
| (26) |                                                                                  |             | MCWB   | 4.5   | 6.3   | 11.5  | 14.6  | 19.1  | 22.9  | 23.7  | 22.4  | 20.7  | 15.4  | 10.3  | 5.7  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.1  | 13.7  | 18.2  | 19.2  | 23.1  | 26.2  | 27.0  | 25.8  | 24.3  | 21.7  | 16.9  | 13.7 | (27) |
| (28) |                                                                                  |             | MCD B  | 14.6  | 17.8  | 25.7  | 27.1  | 29.2  | 32.3  | 33.6  | 32.6  | 32.1  | 27.1  | 22.3  | 16.7 | (28) |
| (29) |                                                                                  | 2%          | WB     | 8.1   | 11.9  | 16.5  | 18.1  | 21.8  | 25.3  | 25.7  | 25.1  | 23.5  | 20.0  | 14.8  | 10.1 | (29) |
| (30) |                                                                                  |             | MCD B  | 13.1  | 16.4  | 21.6  | 25.1  | 27.7  | 32.3  | 32.6  | 32.1  | 31.0  | 25.0  | 19.2  | 13.5 | (30) |
| (31) |                                                                                  | 5%          | WB     | 6.3   | 9.1   | 14.5  | 16.9  | 20.9  | 24.3  | 25.0  | 24.2  | 22.7  | 18.2  | 12.9  | 7.4  | (31) |
| (32) |                                                                                  |             | MCD B  | 10.8  | 15.6  | 20.6  | 23.6  | 27.2  | 31.1  | 32.2  | 31.3  | 29.8  | 23.2  | 18.6  | 11.6 | (32) |
| (33) |                                                                                  | 10%         | WB     | 4.2   | 6.3   | 12.1  | 15.5  | 20.1  | 23.6  | 24.2  | 23.5  | 21.8  | 16.4  | 11.0  | 5.7  | (33) |
| (34) |                                                                                  |             | MCD B  | 8.7   | 11.4  | 17.9  | 21.3  | 26.0  | 30.3  | 31.3  | 29.9  | 28.1  | 21.4  | 15.6  | 9.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 11.5   | 11.9  | 12.5  | 12.6  | 11.7  | 11.5  | 11.6  | 11.8  | 12.5  | 13.3  | 12.4  | 10.8  | (35) |      |
| (36) |                                                                                  | 5% DB       | MCD BR | 15.8  | 16.8  | 16.4  | 15.2  | 13.5  | 12.7  | 13.4  | 13.7  | 12.6  | 15.3  | 15.7  | 14.2 | (36) |
| (37) |                                                                                  |             | MCWBR  | 10.0  | 10.8  | 8.9   | 7.8   | 5.9   | 5.3   | 4.3   | 4.9   | 5.1   | 7.7   | 8.6   | 9.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 15.2  | 15.9  | 14.2  | 13.5  | 11.7  | 11.9  | 11.3  | 11.6  | 11.6  | 13.0  | 14.0  | 12.9 | (38) |
| (39) |                                                                                  |             | MCWBR  | 10.0  | 10.8  | 7.7   | 7.7   | 5.9   | 5.9   | 4.8   | 5.1   | 5.1   | 8.0   | 9.1   | 9.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.286  | 0.301 | 0.332 | 0.378 | 0.408 | 0.420 | 0.426 | 0.425 | 0.381 | 0.341 | 0.300 | 0.281 | (40) |      |
| (41) |                                                                                  | taud        | 2.486  | 2.453 | 2.413 | 2.309 | 2.291 | 2.292 | 2.299 | 2.301 | 2.376 | 2.468 | 2.514 | 2.503 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 897    | 927   | 927   | 898   | 873   | 860   | 851   | 843   | 862   | 868   | 872   | 876   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 77     | 92    | 106   | 126   | 131   | 131   | 129   | 126   | 110   | 90    | 75    | 71    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.32   | 3.09  | 4.05  | 5.13  | 5.86  | 6.65  | 6.55  | 5.85  | 4.84  | 3.54  | 2.54  | 1.99  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.20   | 0.32  | 0.30  | 0.32  | 0.47  | 0.40  | 0.45  | 0.29  | 0.31  | 0.37  | 0.21  | 0.19  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (28 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page